



CoPrompt: A Contrastive-prompt Tuning Method for Multiparty Dialogue Character Relationship Extraction

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ABSTRACT

Information extraction is a fundamental task in natural language processing, and relation extraction is one of its primary subfields. However, most previous research focuses on extracting relations from standard texts, such as news articles and Wikipedia entries, while overlooking the challenges present in dialogues - sparse details, cross-sentence links, and complex character relationships. To address these challenges, we propose a contrastive-prompt tuning method (CoPrompt) to better capture the relationships between characters in dialogues. Our method improves the performance of the relationship extraction model by constructing positive and negative sample pairs to obtain better embedding representations of relationship features through prompt learning. To evaluate our method, we separately built manual and continuous templates and conducted experiments on the DialogRE and CRECIL datasets. Our method consistently outperformed other methods, notably achieving state-of-the-art results on Chinese datasets(DialogRE_cn and CRECIL), underscoring its robust efficiency in Chinese language relation extraction tasks. We also demonstrate the effectiveness of our method in low-resource scenarios. Our code is available at https://github.com/LIyu810/CoPrompt_main.

CCS CONCEPTS

• **Computing methodologies** → **Artificial intelligence**; • **Natural language processing** → Information extraction.

KEYWORDS

Relation Extraction, Multiparty Dialogue, Prompt-tuning

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1 INTRODUCTION

Information extraction is a crucial task in natural language processing, aimed at extracting specific events or factual information from text. In recent years, character relation extraction based on multi-party dialogue has garnered significant attention. From conference content analysis, script analysis, automatic generation of RPG game scripts, to multi-party service-oriented robots, these applications necessitate the understanding and parsing of complex character relationships and interactions within multi-party dialogues. Dialogue-level character relation extraction forms an essential component of information extraction. Especially with the growing demand and applications of multi-party dialogue systems, character relation extraction based on multi-party dialogue text has become a pivotal research focus.

The goal of character relationship extraction is to derive semantic relationships between entities from unstructured dialogue texts. A fragment of CRECIL training data is provided in Table 1 to illustrate the task at the dialogue level. In this instance, "Speaker1" and "Speaker2" symbolize character entities, while "R1" and "R2" signify distinct relationship types between entities.

Pre-training and fine-tuning of language models (PLMs), popularized post-BERT[10], has become a standard protocol in NLP. It leverages pre-trained models for specific tasks, minimizing the need for extensive labeled datasets. However, this approach can result in catastrophic forgetting[3] and still necessitates a certain data volume. A burgeoning alternative, prompt learning, utilizes natural language prompts within models like GPT-3[1], allowing for task completion with minimal examples. Through templates, it triggers factual knowledge within the PLM, mapping this knowledge to form the answer space, thus presenting an efficient alternative to the traditional fine-tuning method.

Table 1: A Fragment of CRECIL dataset.

Speaker1:小凡啊，你又是勤工俭学，啊，又是帮助人家编书，财路不少嘛，完全有能力拿出那么一点来，在这个商品经济大潮里面学游泳么，这对你将来是很有好处地。	
Speaker1: Xiaofan, you are also work-study, ah, and help people compile books, a lot of money, fully capable of taking out so little to learn to swim in this commodity economy tide, which is very good for your future.	
Speaker2: 行了行了爸，多少钱一股您说话。	
Speaker2: All right, all right, Dad, how much money a share do you talk?	
Speaker1: 啊二十块钱一股，我入了十股，你入多少？唉，可不能买的多呀，现在还是很抢手的哩，我们这是内部调济，啊，总还要给群众留一些嘛"	
Speaker1: Ah, twenty dollars a share. I entered ten shares. How much did you enter? Ah, you can not buy more. It is still very sought-after. We are internal transfer. Ah, we always have to leave some to the masses as well.	
Speaker2: 那也行，我呢也甭太特殊了，嗯，我来.....一股，二十块钱，您数好。	
Speaker2: Well, I don't need to be too special. Well, I'll buy one share for twenty dollars. You count it.	
Argument Pair	Relation type
R1 (Speaker1, 小凡)	per:parents
R2 (小凡, Speaker2)	per:alternate_name

Motivated by this, we attempted to solve the dialogue character relationship extraction problem using the prompt-based fine-tuning approach. However, there are several challenges that restrict it from being directly applicable to this task. Dialogue-level RE is distinguished by low information density, a high proportion of cross-sentence relationships, and difficulty in capturing useful character relationship features. The template design in the RE task must take into account more domain-related knowledge, and there is a larger and more complex label space for RE. To overcome these challenges, we present a contrastive prompt fine-tuning approach to address the dialogue character RE problem in this study. Specifically, by creating positive and negative sample pairs and simultaneously optimizing the loss function, we gain a better embedding representation of the relationship characteristics. Our method's efficiency is shown by a vast number of experiments, and our primary contributions are as follows:

- We apply prompt learning to the dialogue character relationship extraction task and build trainable templates to reduce template construction costs.
- We propose the CoPrompt method, creatively incorporating contrastive learning into prompt learning network training to obtain a better embedding representation of relational features.
- We conducted experiments on the CRECIL and DialogRE public datasets, which corroborated the efficacy of our approach, yielding the best performance on the DialogRE_cn and CRECIL datasets, thereby setting a new benchmark for future work. Moreover, comparative experiments with a small sample dataset further validated the effectiveness of our approach in low-resource scenarios.

2 RELATED WORK

2.1 Relationship Extraction

Similar to traditional document-level relationship extraction, dialogue-level relationship extraction takes long text as input and includes

both intra-sentence and inter-sentence relationships. To facilitate research in this area, Yu et al. [24] introduced DialogRE, the first dataset specifically designed for dialogue text-oriented RE. Subsequently, research into dialogue-level relationship extraction tasks began to emerge. Recently, Jiang et al. [8] proposed the CRECIL dataset, which focuses on relationship extraction in Chinese conversation texts. They also provided a baseline model based on this dataset to support academic research in this field.

At the moment, various artificial intelligence tasks represented by deep learning are flourishing, and scholars have applied deep learning to the task of RE, and research on RE for data in the dialogue domain has yielded specific results. With the proposal and application of classical models such as CNN[11], RNN[25], and LSTM[20], more improved models based on such fundamental methods have been proposed. To improve the performance of RE, Zhou et al.[29] proposed a bidirectional LSTM model to capture long-range dependencies between entity pairs, and Zhang et al.[27] proposed PALSTM to encode global location information. Given the significant advantages of graph models in relational reasoning, they are widely used in RE tasks. Zhang et al.[26] used graph convolutional networks (GCN) to capture entity-type interdependencies. Guo et al.[5] proposed a model based on Attention Guided Graph Convolutional Networks (AGGCN). This model fully utilizes the information in the dependency tree to extract the correlations. GDPNet was proposed by Xue et al.[22] to capture possible relationships between tokens using potential multiple views, then refine this graph to select important words for relationship prediction, and finally connect the improved graph representation with the BERT model-based representation for relationship extraction. The above approach is primarily based on the paradigm of fine-tuning PLM, which has become the industry standard and has significantly improved the performance of a wide range of tasks with a unified model architecture.

2.2 Prompt Learning

Since the introduction of GPT-3 [1], the new paradigm of prompt-based learning has gained significant attention from academics. Prompt-based learning converts the downstream task into a form consistent with the pre-trained language model (PLM) task, allowing for lightweight and efficient fine-tuning of the downstream task to complete the task more efficiently. The prompt function $f_{prompt}(x)$ [14] is the core of the prompt-based learning process, where x refers to different template insertions. The objective of prompt-based learning is to optimize the prompt function’s performance on downstream tasks, making it as efficient as possible.

Manual design of templates is typically based on human knowledge of natural language, producing semantically fluent and efficient templates. Petroni et al. [16] proposed manually creating cloze templates on the Llama dataset to stimulate knowledge in PLM. Brown et al. [1] created hand-crafted prefix prompts for various tasks, such as Q&A, translation, and probing tasks for common sense reasoning. However, while the technique of designing templates by hand is straightforward and works well in handling many tasks with a high degree of accuracy in template design, it has significant drawbacks. Firstly, producing and experimenting with these templates is an art that takes time and practice, especially for complex tasks like semantic parsing [18]. Secondly, even experienced prompt designers may struggle to manually discover the optimal templates [9]. Overall, designing templates by hand is more natural, but it requires more experimentation, experience, and language competence, making it more costly.

In response to these challenges, Liu et al. [15] proposed P-tuning, a simple and effective method that transforms template construction into a continuous parameter optimization problem, successfully achieving automatic template construction. Hambardzumyan et al. [6] introduced the use of continuous templates by suggesting that the continuous template’s shape should match the design of the manual template. Shin et al. [19] proposed a gradient-guided search-based automated technique for providing prompts for various tasks, suggesting that fine-tuning may be replaced as PLM gets more complex and powerful. Zhong et al. [28] used the AutoPrompt method to define templates for the first time and found that manual template initialization provides a better starting point for the search process.

For the relational extraction task, Han et al. [7] proposed the PTR method, which uses logical rules to build prompts with multiple sub-prompts. In contrast to PTR’s predefined rules, Chen et al. [2] propose KnowPrompt, a new knowledge-aware tuning method that uses knowledge injection to build prompts and reduce the labor-intensive prompt engineering. It optimizes virtual type words and answer words to each other in a synergistic manner. However, since the CRECIL dataset only contains character type relations, injecting virtual type words is not directly applicable. Furthermore, we need to create Chinese templates for the context of Chinese dialogues to stimulate model knowledge.

2.3 Few-shot learning

Few-Shot Learning (FSL) aims to make predictions based on a limited number of samples. Current supervised learning-based RE methods require a substantial amount of fully labeled training data,

but in reality, there is often a lack of large amounts of manually labeled data. Although remotely supervised models can obtain a significant amount of data by aligning the knowledge base, they cannot effectively solve the long-tail problem of the sample distribution [21]. To address these issues, we conducted extensive experimental and analytical work in a low-resource scenario to demonstrate the effectiveness of our method.

2.4 Contrastive learning

Contrastive learning is regarded as a self-supervised learning (SSL) method that aims to learn a good representation of the data while data labels are set. Fang et al. [4] proposed the CERT approach, claiming that the BERT task is token-based and that the model incorporating the contrastive learning task performs better when obtaining sentence representation. Since the primary goal is feature extraction, this class of methods is primarily used in natural language understanding (NLU). Contrastive learning has emerged in computer vision and has made some progress. Currently, research on contrastive learning in the field of NLP is quite restricted. Based on previous research on contrastive learning, we believe that incorporating a contrastive learning module can help capture the relationships between characters in conversational texts and improve the model’s natural language understanding.

Compared with the aforementioned methods, our approach has two significant differences. Firstly, most of the previous work has been based on English datasets, while we focus on Chinese conversation-level relationship extraction. Additionally, we create continuous templates tailored to the characteristics of Chinese dialogue text to reduce template construction costs. Secondly, we make full use of the data’s features. We design positive and negative sample pairs using contrastive learning without additional data annotation to fully capture the semantic relationships between character entities. As a result, we have a more accurate representation of relational feature embedding.

3 COPROMPT FINE-TUNING FOR RE

In this section, we introduce a prompt fine-tuning method that incorporates contrastive learning, which we call CoPrompt. Figure 1(b) illustrates how we construct and optimize the CoPrompt method. This method combines a strong foundation in both linguistics and technical expertise.

Consider a dialogue D_{in} consisting of n rounds, denoted as $S_1 : T_1, S_2 : T_2, \dots, S_i : T_i, \dots, S_n : T_n$, along with an unidentified relational entity (A_1, A_2) . Here, A_1 and A_2 are subsequences of D_{in} , where S_i and T_i represent the speaker and dialogue text for the i th round, respectively. In general, relational entities consist of multiple tokens, such as $A_1 = a_1, a_2, \dots, a_{m-1}$, where m denotes the number of tokens in entity A_1 . Our task is to predict the relationship r between A_1 and A_2 given a piece of dialogue and entity pair (A_1, A_2) , where $r \in R$ and R is the set of predefined relationship types in the dataset.

To help the model locate the start of speaking rounds based on speaker parameters and prevent overfitting, we used the method for speaker substitution in the input model proposed by Yu et al. [24]. Given a relationship pair (A_1, A_2) and its corresponding dialogue document $D_{in} = \{S_1 : T_1, S_2 : T_2, \dots, S_i : T_i, \dots, S_n : T_n\}$, we construct

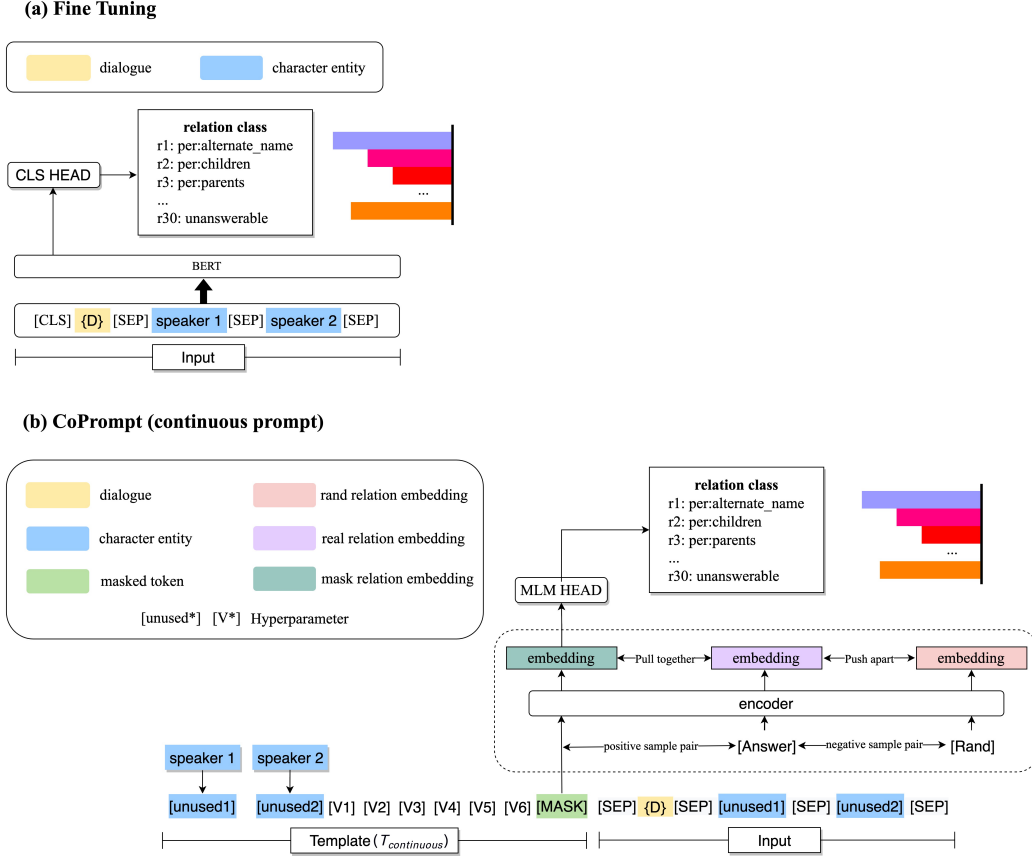


Figure 1: The model architecture of fine-tuning for RE (a), and the proposed CoPrompt (b) approach.

a modified version $D_{in}' = \{S_1' : T_1, S_2' : T_2, \dots, S_i' : T_i, \dots, S_n' : T_n\}$. Specifically, if the entity to be recognized contains a speaker, we replace the speaker's name with a special token $Unused_i$, and modify S_i' according to Equation 1.

$$S_i' = \begin{cases} [Unused1] & \text{if } S_1 = A_1 \\ [Unused2] & \text{if } S_2 = A_2 \\ S_i & \text{otherwise} \end{cases} \quad (1)$$

3.1 Template engineering for multiparty datasets

Templates can serve as probes for language models, allowing us to extract specific knowledge to improve their migration ability. In prompt engineering, the goal is to enable language models to perform tasks efficiently, and it's not necessary for the prompt to be interpretable by humans. Li et al. [13] have shown that using a continuous word embedding as a prompt is more expressive than using discrete word embeddings. To create learnable templates, we added additional tokens ($[V_1] \dots [V_m]$) to the vocabulary when building continuous templates. In continuous template design, the number of tokens m is a hyperparameter, and these additional tokens are randomly initialized and participate in the training process. We inserted the designed template in front of the dialogue text (D_{in})

during training. The trainable template has parameters that can be adjusted based on the downstream task's training data, as shown in Equation 2. To enable a systematic comparison between the continuous template ($T_{continuous}$) and the manual templates (T_{manual}), we devised two types of T_{manual} as references. These specifically designed templates, displayed in Equations 3 and 4, correspond to the templates for the Chinese and English datasets respectively.

$$T_{continuous} = [S_1] [S_2] [V_1] [V_2] \dots [V_m] [MASK] \quad (2)$$

$$T_{manual} = [S_1] \text{和} [S_2] \text{的关系是} [MASK] \quad (3)$$

$$T_{manual} = \text{The relationship between } [S_1] \text{ and } [S_2] \text{ is } [MASK] \quad (4)$$

3.2 Answer Space Design Methods

To generate a valid prediction model, answer engineering searches for the mapping relationship between the answer word space Z and the original output Y . In the relation extraction task, Z represents the answer word space corresponding to the [MASK] token, and Y denotes the relation label. When using prompt learning to solve classification problems, it's common to add a verbalizer to translate

the predictions for words at the [MASK] position into categorical labels. For example, in a text classification task, the following mapping relationship might exist: POLITICS: "politics", SPORTS: "sports". When the masked language model predicts that the word with the highest probability at the [MASK] position is "politics", the sentence is classified as POLITICS. Thus, we can use the PLM score at the [MASK] position for the word "politics" as the predicted score for the POLITICS category.

However, the multiparty relation extraction task involves many categories of character relationships, and it's usually impossible to properly represent a category of character relationships with a single word in the Chinese PLM vocabulary. As a result, we map the answer word space Z directly to the final category label Y .

To better leverage the rich semantic knowledge in relationship labels for relation extraction and to represent the relationships between characters in the word list space with their unique token, we followed the approach presented in KnowPrompt [2] to encode the semantic knowledge of the labels. Specifically, we introduced a virtual answer word set V' in the PLM word list to express the relationship's implicit semantics. The masked language model's head layer (MLM) was extended with additional learnable relational embeddings to fully represent the corresponding relational labels Y .

Since we believe that good initialization is critical when initializing the relation embeddings, we first decomposed the relation labels into their constituent words and then calculated a weighted average of the decomposed tokens to obtain the relation's initialization. For example, to set the *per* : *children* label, we calculated the word embedding weighted average of *per*, *chil*, *dren*.

3.3 Joint Optimization with Contrastive Learning

We propose the CoPrompt method to enhance the performance of multiparty entity relationship extraction models. Our approach introduces positive and negative samples to calibrate the feature representation of the word embedding corresponding to the [MASK] token. By focusing on the more relevant relational characteristics of the dialogue, CoPrompt improves model performance without requiring additional data annotation. Specifically, we use the answer labels as anchors, predicted relationship labels as positive samples, and randomly chosen incorrect relationship labels from the relationship list as negative samples. Our objective is to reduce the distance between positive samples and anchor samples while increasing the distance between negative samples and anchor samples in the vector representation space. To achieve this, we define the loss function L_c as shown in Equation 5.

$$L_c = -\log\sigma(\cos(\text{real}, \text{mask}) - \gamma) - \log\sigma(\gamma - \cos(\text{real}, \text{rand})) \quad (5)$$

In the equation, σ denotes the sigmoid function, *real* represents the anchor sample, *mask* represents the positive sample, and *rand* represents the negative sample. We chose the cosine similarity as the scoring function (as shown in Equation 6) to measure the distance between positive and negative sample pairs in the projection space. We used the contrastive loss L_c to optimize the feature representation of the word embeddings corresponding to the [MASK]

token and the loss function L_m to fine-tune the parameters of the pre-trained language model. The term "CE" in the formula represents the cross-entropy loss function (as shown in Equation 7), and λ denotes the weight of the contrastive loss L_c . Finally, we obtain the training loss as shown in Equation 8.

$$\cos(u, v) = \frac{\langle u, v \rangle}{\|u\| \|v\|} \quad (6)$$

$$L_m = CE(P(y|X_{\text{prompt}})) \quad (7)$$

$$L = L_m + \lambda L_c \quad (8)$$

4 EXPERIMENTS

In this section, we conduct extensive experiments on the conversation-level relation extraction task to demonstrate the effectiveness of our approach, and the experiments are evaluated using the F1 score for the recognition results.

4.1 RE Scenario

In the fully-supervised setting, CoPrompt is evaluated against both fine-tuning-based and prompt-based fine-tuning methodologies. For the fine-tuning based approaches, we consider models such as BERT [9], BERT_S [24], GDPNet [22], TUCORE-GCN [12], SimpleRE [23] and SocAog[17]. As for the prompt-based fine-tuning approaches, we consider the KnowPrompt [2] as the primary contender. These models were selected based on their prior demonstrated effectiveness in the relationship extraction tasks, with diverse methodologies that provide a comprehensive assessment of our approach.

In the low-resource setting, we cannot randomly select samples based solely on relationship labels, as the CRECIL dataset is organized into dialogue scenes. Therefore, we take all the relationship pairs identified within a conversation scene as a single sample. However, this approach leads to incomplete coverage of relationship categories in the training set. To address the issue of unstable experimental results in few-shot learning, we use a fixed set of random seeds for sampling and average the experimental results after five sets of experiments as the final result. To gain a more in-depth analysis of model performance, we also conduct experiments with 8, 16, 32, and 64 samples. Table 2 shows the data distribution taken five times using random seeds in the 8-shot experiment.

4.2 Datasets

We conducted experiments on two RE datasets: DialogRE¹, proposed by Yu et al., and the CRECIL² dataset, proposed by Jiang et al.

The CRECIL dataset collects character relationships from the Chinese sitcom "I Love My Family," which contains Chinese dialogues of complex family character relationships. In 686 dialogue scenes, the relationships between 140 character entities were manually labeled, covering 30 relationship categories. Table 3 shows the detailed structure of the dataset, and the number of instances indicates the number of entity pairs to be identified. In the CRECIL

¹<https://github.com/nlpdata/dialogre>

²<https://github.com/bistu-nlp-lab/CRECIL.git>

Table 2: 8-shot Statistics.

Sampling	Examples	Number of relations
8-6	1102	20
8-7	702	21
8-8	992	21
8-9	502	15
8-10	902	18

dataset, the average number of rounds per conversation is 29.4, and the average number of participants is 4.1. The more multi-party conversations in the dataset, the more difficult it is for the model to extract the exact character relationships.

DialogRE, the pioneering dialogue-level RE dataset, comprises manually annotated task dialogues sourced from the "Old Friends" show. It includes 1788 dialogues with 10,168 entity pairs, spanning 37 distinct relationship types. Available in both English and Chinese, this dataset has been utilized for our evaluation to afford a comprehensive assessment of our model across multiple languages. The detailed structure of the DialogRE is presented in Table 4.

Table 3: CRECIL dataset structure.

	Train	Test	Dev
examples	40454	7812	5380
number of relations	30	30	30

Table 4: DialogRE dataset structure.

	Train	Test	Dev
examples	6660	2125	2102
number of relations	35	37	35

4.3 Experimental Settings

The experiments were conducted using the PyTorch 1.8.1 framework and accelerated with an NVIDIA 3090 GPU. We utilized the *Roberta_{base}* pre-training model with the embedding dimension set to 768 and the maximum sequence length set to 512. The gradient accumulation step was set to 3, and we used an Adam optimizer with an early stopping parameter of 20 epochs. The batch size was set to 6, with a L_c learning rate of $2e-5$ and a L_m learning rate of $5e-5$. We set λ to 0.01 and γ to 0.3.

4.4 Main Results

4.4.1 Full-Supervised RE. As illustrated in Table 5, we compared fine-tuning based methods, prompt-based fine-tuning methods, and our CoPrompt approach in a fully supervised setting. While some fine-tuning based models like TUCORE-GCN performed well on

specific datasets (e.g., *DialogRE_en*), their performance was variable across datasets. Alternatively, the prompt-based method, Know-Prompt, delivered stable results, albeit slightly lower than the best-performing fine-tuning methods. Our CoPrompt approach demonstrated exceptional performance, with our *CoPrompt_{continuous}* method achieving high F1 scores of 61.9% and 69.1% on the CRECIL and *DialogRE_cn* datasets, respectively. This superior performance, particularly on Chinese language datasets, underscores the effectiveness of our approach. However, while *CoPrompt_{continuous}* exhibited overall strong performance, it was not able to surpass the current best model, SocAog, on the *DialogRE_en* dataset. This indicates potential areas for future improvements to further optimize our model's performance across all contexts.

4.4.2 Few Shot RE. We conducted experiments on the CRECIL dataset in a low-resource scenario, and the results are presented in Table 6. As it currently stands, the standard fine-tuning model is the sole research method that has been applied to this dataset. Therefore, in our comparative analysis, we contrasted our proposed method, CoPrompt, with the existing standard fine-tuning model. The CoPrompt method significantly outperformed the standard fine-tuning model in scenarios with fewer samples, especially in the 64-shot experiment. With continuous templates, our method improved by 2.3 percent compared to standard fine-tuning in the 8-shot setting. As the number of conversation samples increased to 64, our method improved by 8.2 percent.

Based on the experimental results, we conclude that the continuous template is preferable for Chinese relationship extraction as it eliminates the need for expert knowledge and can lead to more efficient optimization and better predictions. Our CoPrompt method achieves better performance on both the *CRECIL* and *DialogRE_cn* datasets, demonstrating that using positive and negative sample pairs to calibrate feature representation of word embeddings corresponding to [MASK] can improve model efficiency. CoPrompt is a simple and effective paradigm for fine-tuning Chinese RE, which can be applied to the Chinese relational extraction task. Furthermore, the performance gains of CoPrompt steadily increase from 8 to 64 samples compared to standard fine-tuning. As shown in Figure 2, having more data can significantly improve model performance in low-resource scenarios.

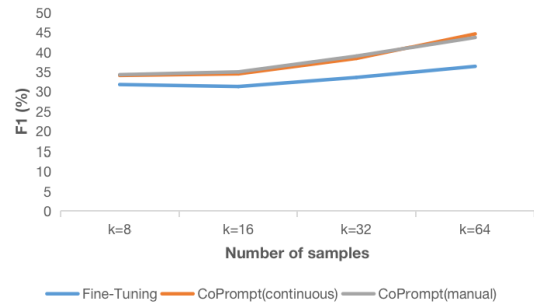


Figure 2: In the low-resource scenario, the growth curves correspond to the F1 values of various models as the number of samples increases.

Table 5: Standard RE performance of F_1 scores (%) on CRECIL, DialogRE_cn, and DialogRE_en test sets. Best results are bold.

Full-Supervised Setting			
Methods	CRECIL	DialogRE_cn	DialogRE_en
Fine-tuning based approach			
<i>BERT</i> [9]	54.4	61.6	57.9
<i>BERT_S</i> [24]	-	-	59.5
<i>GDPNet</i> [22]	-	62.2	60.2
<i>TUCORE-GCN</i> [12]	-	-	65.5
<i>SimpleRE</i> [23]	-	65.2	66.7
<i>SocAog</i> [17]	-	-	69.1
Prompt-based fine-tuning approach			
<i>KnowPrompt</i> [2]	54.4 [*]	-	66.8 [*]
<i>CoPrompt_{manual}</i> (Ours)	58.7	67.8	65.9
<i>CoPrompt_{continuous}</i> (Ours)	61.9	69.1	67.4

Table 6: Low-resource RE performance of F_1 scores (%) on CRECIL test set. We use K = 8, 16, 32,64 (examples per dialogue) for few-shot experiments. Best results are bold.

Low-Resource Setting								
Methods	k=8		K=16		K=32		K=64	
	Dev	Test	Dev	Test	Dev	Test	Dev	Test
<i>Fine-Tuning</i> [9]	27.9	31.8	29.7	31.3	32.2	33.6	36.8	36.4
<i>CoPrompt_{manual}</i> (<i>Ours</i>)	32.0	34.3	33.2	35.0	39.4	39.0	47.0	43.7
<i>CoPrompt_{continuous}</i> (<i>Ours</i>)	30.3	34.1	31.6	34.5	38.7	38.4	46.3	44.6

4.4.3 Evaluation of Large-Scale Models. With the surge in development of large-scale models, we assessed the performance of popular models such as ChatGPT 3.5 and chatglm on the CRECIL dataset. Notably, despite their outstanding performance in many NLP tasks, these models fell short in our relation extraction task. Recognizing the crucial role of prompts in harnessing the potential of large-scale models, we created a series of purpose-designed prompts aiming to maximize their effectiveness. However, the outcomes were less than satisfactory. For instance, the precision of ChatGPT 3.5 was approximately 2%, while chatglm-6b achieved around 20% accuracy. These results suggest that even though prompt design can be beneficial, it does not inherently guarantee superior results in specific tasks such as relation extraction. The performance of these large-scale models was significantly lower than that of CoPrompt, further reinforcing that while large models can handle generic NLP tasks effectively, their applicability is somewhat limited in more specialized domains. These observations underscore the value and efficacy of our proposed CoPrompt method, specifically tailored to address the complexities of relation extraction from dialogues.

4.5 Visualization

To delve deeper into how CoPrompt enhances the quality of representation, we utilized t-SNE plots to depict the learned representations on the CRECIL test set. We selected three character relationship types with the highest coverage in the CRECIL dataset for visual display: per:neighbor (represented in blue), per:children

(in green), and per:parents (in red). Figure 3 illustrates the results of the fine-tuning, where the sample distribution manifests a uniform and random pattern. However, Figure 4 portrays the sample distribution derived from the method without the incorporation of contrastive loss optimization. Lastly, the CoPrompt method is demonstrated in Figure 5. Evidently, CoPrompt method results in a more compact and focused sample distribution, akin to clustering effects. Observing the evolution from Figure 3 to Figure 5, we can perceive the gradual improvement in clustering effects. This strongly implies that in dealing with complex character relationship representation learning tasks, the CoPrompt method can effectively enhance the quality and robustness of classification, thus realizing an efficient classification of samples.

5 ABLATION STUDY ON COPROMPT

As previously discussed, extensive experiments demonstrate the effectiveness of the CoPrompt method. In this section, we conduct additional ablation studies on the CRECIL dataset to evaluate the significance of incorporating contrastive learning modules.

Table 7 displays the experimental results for manual and continuous prompts on the CRECIL dataset. For *CoPrompt* - L_c , we take the output features of [MASK] to be predicted directly by the MLM Head, without participating in optimizing the contrastive loss L_c . This considerably reduces the model’s classification effectiveness in the fully supervised scenario. In the scenarios of continuous and manual prompts, the F1 values of the test set decrease by 3.2 and

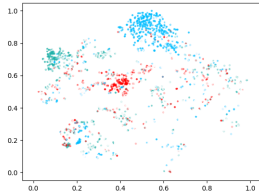


Figure 3: Fine-Tuning

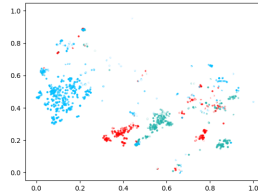
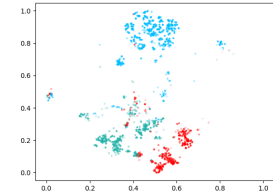
Figure 4: CoPrompt- L_c 

Figure 5: CoPrompt

3.5 percent respectively, when tested on the CRECIL dataset. The addition of contrastive learning significantly improves the model’s effectiveness, as evidenced by the results from the CRECIL dataset.

Table 7: Ablation Study on CoPrompt over the CRECIL Dataset. Subscript in red represents the advantages of CoPrompt over the results of ablation study.

Model	F1(%)
$CoPrompt_{manual} - L_c$	55.2 (-3.5)
$CoPrompt_{continuous} - L_c$	58.7 (-3.2)

6 CONCLUSION AND FUTURE WORK

In this paper, we propose the CoPrompt method, which is a simple and effective language model fine-tuning paradigm applied to the multiparty relation extraction task. We further reduce the template construction cost by designing trainable templates. Additionally, we use contrastive learning without the need for additional data annotation. Designing positive and negative sample pairs to capture the semantic relationships between role entities yields a better representation of relational feature embedding. Experiments demonstrate that our approach significantly outperforms standard fine-tuning in fully supervised and few-shot scenarios. Our future work will focus on sophisticated conversation-level relationship extraction tasks and low-resource prompt tuning methods.

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